Posterior Distortion from Hypothesis-Dependent Validity Gating in Runtime Diagnosis

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Abstract

A runtime diagnostic layer decides, tick by tick, which of its evidence is *applicable*. That decision is a coarsening mechanism the system designed for itself, and whether it may be ignored is a property of the software rather than of the data. We give the exact algebra. Writing q for the posterior the system would have obtained from *all* the evidence it could in principle have scored and p for the posterior it actually produces, $p_h = q_h w_h / \mathbb{E}_q[w]$, so every pairwise log-posterior ratio acquires an additive, evidence-independent offset $\log(w_a/w_b)$ that further data of the same kind cannot argue with. For the additive gated scores runtime layers compute, w_h is exhibited in closed form and the offset is exactly the withheld distinguishing contribution. When no active term distinguishes two hypotheses their ratio is pinned to the prior ratio exactly, for as long as the gate stays shut, however concentrated the posterior becomes — a statement about the produced posterior alone, which does not depend on the choice of baseline. We give the condition under which the distortion vanishes, its multivalued caveat, a sufficient condition for the MAP to be unchanged, and exact divergence identities: $\mathrm{KL}(q\|p)$ is precisely the Jensen gap of $\log w$ under q, so the distortion is zero exactly when the validity factors do not vary across hypotheses.

We claim no new mathematics. Each result is an instance of something known — self-normalised reweighting, argmax stability under bounded perturbation, Jensen's inequality — and the setting is the coarse-data one, so the paper concedes rather than claims at every step. What we offer is the runtime-diagnostic formulation, the observation that the condition is decidable *statically, from a signature table, before the system runs*, and a deployed layer that fails it: an independently written checker that imports none of the production analysis agrees with it on 378,000 of 378,000 tick-by-pair comparisons and finds one hypothesis pair pinned to its prior for 200 consecutive ticks.

1 The problem

Diagnostic systems that run online do not evaluate every test at every instant. A test whose preconditions are unmet is *not applicable*, and a common and natural implementation is to skip it: the corresponding term is removed from the score. This is not a bug. Scoring a test whose preconditions are unmet would inject noise, and down-weighting it requires a weight nobody has measured.

The difficulty is that applicability is often *hypothesis-relative*, and that the condition producing the fault may be the same condition that removes the applicability. A receiver that loses its fix stops emitting the observable that would have named the loss. The diagnostic layer is then reasoning about a hypothesis using evidence the hypothesis itself destroyed.

Two questions follow, and this paper answers them exactly.

1. How far can the produced posterior be from the posterior the surviving evidence supports, and in what direction?

2. When is the difference nil, and is that condition checkable?

What this paper does not do. It makes no observability claim: the physical fault may be perfectly detectable by other means, and in the system we measure it is. It offers no new inequality, no new estimator and no calibration. It does not show that any system was unsafe.

2 Setting

Let $H = \{h_1, \ldots, h_n\}$, $n \geq 2$, be a finite set of mutually exclusive hypotheses, π_h the prior, L_h a likelihood under h, and $w_h > 0$ a hypothesis-dependent validity factor. Define

$$q_h \;=\; \frac{\pi_h L_h}{\sum_j \pi_j L_j}, \qquad p_h \;=\; \frac{\pi_h w_h L_h}{\sum_j \pi_j w_j L_j}. \tag{1}$$

Assumption 1 (strict positivity)**.** $\pi_h, L_h, w_h \in (0, \infty)$ for all h.

A closed gate does not violate Assumption 1. The reader has just met the word *skip*, and it is natural to read a skipped feature as $w_h = 0$. It is not: by (7) a shut gate gives $w_h = \exp(-\ell_j(h))$, strictly positive and finite for any finite $\ell_j(h)$. What Assumption 1 excludes is a zero *likelihood* — the *exclusion* of a hypothesis, a different operation with different semantics, and out of scope here.

Assumption 2 (finiteness)**.** $n < \infty$.

Assumption 3 (no independence is assumed)**.** Nothing below assumes the gate is independent of the residual, or of the fault. The gate may be an arbitrary function of the same state the evidence is a function of.

Assumption 4 (w is exogenous to the algebra)**.** w is whatever the implementation makes it. We say what follows *given* w; we do not model how w arises, except in Section 5 where a structural condition on it is stated explicitly.

Assumption 3 is worth dwelling on. A reader familiar with missing-data arguments will expect an independence or ignorability condition to be doing the work somewhere. It is not. The identities below are unconditional; ignorability appears only in Section 5, as the special case in which the distortion vanishes.

2.1 What q is measured against, and why it matters

Everything below compares two posteriors, so the *baseline* q must be named. There are two candidates and they are not interchangeable.

Reading A — active-evidence baseline. L_h is the likelihood of the evidence that survived the gate. Then the gating is already inside L_h, there is nothing left for w to carry, $w_h \equiv 1$, and $p = q$ identically.

Reading B — full-evidence baseline. L_h is the likelihood of *all* the evidence the system could in principle have used, and w_h carries what the gate withheld. Then $p \neq q$ in general.

Reading A makes every result in this paper vacuous, and we say so before a reader discovers it. Under Reading A, Propositions 1, 2, 9 and 10 are true and empty: they describe a reweighting by the constant 1. An earlier draft of this paper defined L_h that way and presented Section 10 as an instance of the framework; it was not one.

Everything below uses Reading B, and Proposition 4 makes it concrete for the additive scores that runtime diagnostic layers actually compute — including the one in Section 10, whose w we exhibit in closed form.

Remark 1 (q is counterfactual, and the main claim does not need it)**.** Under Reading B the baseline q is generally *not computable at runtime*, and in the case Section 10 measures it is not computable at all: the gate closes because the receiver has stopped reporting, so the withheld $\ell_j(h)$ has no observed value to take. q is a reference point, not a quantity we claim to evaluate, and w_h is the exact bookkeeping of what separates the two — no more than that.

This matters less than it appears, because **the paper's empirical claim does not mention** q. Proposition 3 and Corollary 5 relate p to the *prior*, both of which the system computes. A reader who rejects Reading B entirely still owes an answer to Section 10, where a posterior ratio sat at its prior value for 200 consecutive ticks. What such a reader loses is the quantitative apparatus — Propositions 9 and 10 measure a distance to a baseline, and a baseline that cannot be evaluated cannot be measured against.

Write $\mathbb{E}_q[w] = \sum_h q_h w_h$, $w_{\max} = \max_h w_h$, $w_{\min} = \min_h w_h$, and

$$R \;=\; \log \frac{w_{\max}}{w_{\min}} \;\geq\; 0, \tag{2}$$

the *log range* of the validity factors.

3 The reweighting identity

Proposition 1 (reweighting)**.** *Under Assumptions 1–2, for every h,*

$$p_h \;=\; q_h \, \frac{w_h}{\mathbb{E}_q[w]}. \tag{3}$$

Proof. Let $Z_q = \sum_j \pi_j L_j$ and $Z_p = \sum_j \pi_j w_j L_j$, both finite and positive. Then $q_h w_h / \mathbb{E}_q[w] = (\pi_h L_h / Z_q)\, w_h / \sum_j (\pi_j L_j / Z_q) w_j = \pi_h w_h L_h / Z_p = p_h$. □

The normalising constants cancel, and that cancellation is the whole content: p is the *self-normalised* reweighting of q by w.

Proposition 2 (log-ratio decomposition)**.** *For any $a, b \in H$,*

$$\log \frac{p_a}{p_b} \;=\; \log \frac{q_a}{q_b} \;+\; \log \frac{w_a}{w_b}. \tag{4}$$

Proof. By (3), $p_a/p_b = (q_a/q_b)(w_a/w_b)$; $\mathbb{E}_q[w]$ cancels. □

Equation (4) is the object the rest of the paper is about. The gating contributes an **additive, evidence-independent** shift to every pairwise log-posterior ratio. It does not scale the evidence, does not interact with it, and does not diminish as evidence accumulates. $\log(w_a/w_b)$ is a constant offset the data cannot argue with.

4 Pinning, and the condition that causes it

Proposition 3 (pinned to the prior)**.** *If $L_a = L_b$ and $w_a = w_b$, then $p_a/p_b = \pi_a/\pi_b$ exactly, for every value of the observed evidence.*

Proof. $q_a/q_b = \pi_a/\pi_b$ since $L_a = L_b$; apply (4) with $\log(w_a/w_b) = 0$. □

Proposition 3 is stronger than "the evidence is weak". The ratio is not merely slow to move; it *cannot* move. No quantity of further evidence of this kind changes it.

When does $L_a = L_b$ hold in practice? For an additive score over per-feature terms — the naive-Bayes signature form used by the system in Section 10, and by many runtime diagnostic layers — write $\ell_j(h)$ for feature j's log-likelihood contribution under h and $g_j \geq 0$ for its gate. Then

$$\log L_a - \log L_b \;=\; \sum_{j \in \mathrm{ACTIVE}} \big(\ell_j(a) - \ell_j(b)\big), \qquad \mathrm{ACTIVE} = \{j : g_j > 0\}. \tag{5}$$

Every feature contributing identically to a and b vanishes from this sum identically, for every value of the observation. Define the *active distinguishing set*

$$D(a,b) \;=\; \{\, j \in \mathrm{ACTIVE} : \ell_j(a) \neq \ell_j(b) \,\}. \tag{6}$$

If $D(a,b) = \emptyset$ then $L_a = L_b$.

4.1 The gated-score form, which is what runtime layers compute

The abstract w_h of Section 2 is not something an implementer recognises. This makes it concrete, and closes the gap Section 2.1 opened.

Proposition 4 (gated score)**.** *Let the score be additive over per-feature terms,* $\log L_h^{\mathrm{gated}} = \sum_j g_j\, \ell_j(h)$ *with gates* $g_j \in [0,1]$*, and let* q *be the posterior from the ungated score* ($g_j \equiv 1$*, Reading B). Then* p *is the reweighting of* q *by*

$$w_h \;=\; \exp\Big(-\sum_j (1 - g_j)\, \ell_j(h) \Big), \tag{7}$$

Proposition 1 applies, and for any pair

$$\log \frac{p_a}{p_b} - \log \frac{q_a}{q_b} \;=\; -\sum_j (1 - g_j)\big(\ell_j(a) - \ell_j(b)\big). \tag{8}$$

Proof. $\log L_h^{\mathrm{gated}} = \sum_j \ell_j(h) - \sum_j (1-g_j)\ell_j(h) = \log L_h + \log w_h$ with w_h as in (7), which is strictly positive and finite. Apply Proposition 1, then Proposition 2. □

Equation (8) says the shift is **exactly the withheld distinguishing contribution**: a fully open gate contributes all of its term, a fully closed one contributes none, and a partial gate contributes in proportion. And w_h is hypothesis-dependent precisely when some gated feature contributes differently to two hypotheses — which is the ordinary case, not a contrived one.

Corollary 5 (Proposition 3, recovered)**.** *If* $g_j = 0$ *for every* j *with* $\ell_j(a) \neq \ell_j(b)$*, then* $\log(p_a/p_b) = \log(\pi_a/\pi_b)$.

Proof. By (8), $\log(p_a/p_b) = \log(\pi_a/\pi_b) + \sum_j g_j(\ell_j(a) - \ell_j(b))$, and every surviving summand has $\ell_j(a) = \ell_j(b)$. □

So Proposition 3 is not a separate result: it is the corner of Proposition 2 in which the gate withholds every distinguishing term. We state it separately only because it is the case Section 10 measures.

The static-checkability observation. For a signature model in which h is characterised by the feature set $\mathrm{LOW}(h)$ it drives low, $\ell_j(a) \neq \ell_j(b)$ exactly when j lies in the symmetric difference, so

$$D(a,b) \;=\; \big(\mathrm{LOW}(a) \,\triangle\, \mathrm{LOW}(b)\big) \;\cap\; \mathrm{ACTIVE}. \tag{9}$$

The first factor is a property of the *signature table* and can be computed before the system runs. The second is the gate state. So the question "can this pair ever be pinned?" is answered by a set operation on a configuration file, and the question "is it pinned now?" costs one intersection. This is the observation we regard as the paper's practical contribution, and Section 10 is what it buys.

5 Ignorability, and a caveat that is easy to get wrong

Proposition 6 (constant validity factor)**.** *If $w_h = c$ for all h and some $c > 0$, then $p = q$ identically.*

Proof. $\mathbb{E}_q[w] = c$, so $p_h = q_h c/c = q_h$ by (3). □

Remark 2 (what p is, and what "distortion" does not mean)**.** For binary gates and a score additive over features that the model treats as conditionally independent given h — the naive-Bayes form runtime layers use, and the form assumed by (7) — the surviving terms multiply to the marginal likelihood of the surviving features:

$$\prod_{j \in \text{ACTIVE}} \exp \ell_j(h) \;=\; \mathbb{P}\big(x_{\text{ACTIVE}} \mid h\big), \qquad \text{so} \qquad p_h \;\propto\; \pi_h \, \mathbb{P}\big(x_{\text{ACTIVE}} \mid h\big). \tag{10}$$

So p is the posterior given the evidence that survived. If the gate were ignorable, p would be the right answer to a smaller question rather than a wrong answer to the original one, and the gap between p and q would be information loss rather than error. We state this at the outset because it is the first objection a missing-data reader will raise and because, under ignorability, the objection is correct.

It is not the situation of Section 10, for the reason in Remark 3. Throughout this paper "distortion" names the multiplicative reweighting of (3) and nothing more. The propositions are identities; whether a particular reweighting is a defect is a question about the mechanism, and it is answered in the next remark rather than by the algebra.

Remark 3 (the relation to coarsening at random, stated exactly)**.** The gate is a coarsening mechanism, and the framework that owns it is coarse-data ignorability [4]; the same setting is examined, in a stochastic and estimated form, as state-dependent missingness in latent-state models [7]. We claim neither and cite both.

The relation has to be stated carefully, because it is easy — and an earlier draft of this paper did it — to call Proposition 6 "the ignorability condition". It is not. Coarsening at random is a condition on the *mechanism*: the probability of the observed coarsening must not depend on what was coarsened away. What it licenses is dropping the mechanism from the likelihood, leaving exactly the $\mathbb{P}(x_{\text{ACTIVE}} \mid h)$ of (10) — *not* $p = q$. Proposition 6 is a condition on the *withheld evidence*: w constant means nothing discriminating was withheld, and that does give $p = q$. The second is strictly stronger, and the two are independent in both directions. A hypothesis-blind gate that happens to remove a discriminating feature satisfies the first and not the second; a gate that withholds only non-discriminating features while being triggered by the fault satisfies the second and not the first.

For the layer of Section 10 the mechanism condition fails, and it fails for the reason the missing-data literature exists to name: the gate closes *because* the sensor stopped reporting, so the event $\{\text{ACTIVE} = A\}$ itself carries information about the hypothesis. A posterior that used that information would carry a factor $\mathbb{P}(\text{ACTIVE} = A \mid h)$. The implementation carries no such factor — a gated feature contributes nothing to the score and the prior is fixed — so the produced p is neither q nor the posterior that accounts for the gate. Supplying the missing factor would require a per-hypothesis model of gate closure that the system does not have and that we have not validated; we identify the omission and do not repair it here.

What the paper adds is the reading rather than either condition: because the coarsening mechanism is a component of the diagnostic software, both questions above are settled by inspecting a configuration file, before the system runs and without data.

The caveat. The condition is that w be *constant across hypotheses*, not that the gate be random, and not that the gate be conditionally independent of the fault. With a binary gate

applied uniformly to all hypotheses, the surviving-evidence factorisation yields a single shared w and Proposition 6 applies. With a *multivalued or partial* gate — $g_j \in [0,1]$, as real implementations often have — the per-feature contribution is $g_j\,\ell_j(h)$, and the hypothesis dependence re-enters through ℓ_j even when g_j itself is hypothesis-blind. Constancy of w then requires the gate-weighted likelihood difference to vanish, which is Proposition 3's condition and not an independence statement.

This is not hypothetical for the layer in Section 10. Equation (7) gives $w_a/w_b = \exp(-S)$ with $S = \sum_j(\ell_j(a) - \ell_j(b))$ over the withheld features, so $w_a \neq w_b$ exactly when the gate withholds something that separates the pair — which is the case we measure. The ignorability condition fails computably, not merely in principle.

Remark 4. Conditional independence of the gate from the fault does not by itself give $p = q$ in the multivalued case. We state this because the binary intuition is strong and the multivalued case is the common one.

6 Concentration is not discriminating support

Corollary 7. *If $D(a,b) = \emptyset$ — equivalently, if the gate withholds every feature that distinguishes a from b — then $\log(p_a/p_b) = \log(\pi_a/\pi_b)$ regardless of $\max_h p_h$. In particular p_a may be arbitrarily close to 1 — driven by the prior and by likelihood terms that do not distinguish a from b — while the evidence contains nothing separating them.*

Proof. Corollary 5; the conclusion involves no property of $\max_h p_h$. □

Remark 5 (why this is stated through D and not through w)**.** In the abstract setting, pinning is Proposition 3: $L_a = L_b$ *and* $w_a = w_b$. Under Reading B that pair of hypotheses is the wrong one to reach for, because a gate that withholds a distinguishing feature makes $L_a \neq L_b$ and $w_a \neq w_b$ *both* — and the ratio is pinned anyway, because by (7) the two inequalities cancel exactly. The gate-level statement, Corollary 5, is the one that holds in the implemented case; Proposition 3 is its abstract shadow.

Two different quantities are in play and an implementation that reports only one of them will mislead its consumer:

posterior concentration	a function of all hypotheses' scores and of the prior
discriminating support	a function of which terms are active

Remark 6 (terminology)**.** We do not call the posterior a *confidence*. A value of 0.9998 with $D(a,b) = \emptyset$ carries no information about a versus b, and the word invites a reader to think it does.

Corollary 7 says what *cannot* be concluded. It does not say a and b are indistinguishable in principle: another sensor, another relation, or an ungated version of the same relation may separate them. The distinction matters practically, because it points at instrumentation rather than at inference.

7 MAP robustness

Proposition 8 (sufficient condition)**.** *Let $\Delta = \log q_{(1)} - \log q_{(2)}$ be the log margin between the top two hypotheses under q. If $\Delta > R$ then $\arg\max_h p_h = \arg\max_h q_h$.*

Proof. By (3), $\log p_h = \log q_h + \log w_h - \log \mathbb{E}_q[w]$, and the last term is common to all h; ranking under p is the ranking of $\log q_h + \log w_h$. For the q-argmax a and any $b \neq a$, $(\log q_a + \log w_a) - (\log q_b + \log w_b) \geq \Delta - R > 0$. □

Remark 7 (sufficient, not necessary)**.** Over 20,000 random instances the condition held in 2,721, and in every one of those the argmax agreed. In 6,877 instances the argmax moved *while the condition did not hold*, which is permitted. We report the second number so the proposition is not read as a characterisation.

Proposition 8 is an instance of the standard argument that an optimiser is unchanged when the decision margin exceeds the perturbation bound; we claim it as known.

Remark 8 (the proposition is silent exactly where the paper is loudest)**.** R is the spread of the withheld log-likelihood across hypotheses, so it is *largest* in the regime Section 10 measures. Sharply: let a, b be a pinned pair carrying equal prior mass, and suppose they are the top two under q. Writing $S = \sum_j (\ell_j(a) - \ell_j(b))$, (7) gives $\log w_a - \log w_b = -S$, so $R \geq |S|$, while $\Delta = |\log(\pi_a/\pi_b) + S| = |S|$. Hence $\Delta \leq R$ *always*, with equality when the pair attains the extremes of w: **Proposition 8 can never certify a pinned pair.** The audit measures this over the 2,282 sampled instances in which such a pair was the top two and finds a worst slack of $+8.88 \times 10^{-16}$, equality to floating-point precision and no violation.

The equal-prior hypothesis is doing real work and there is an escape, which is worth naming because it makes the position worse rather than better. Since $\Delta \leq |\log(\pi_a/\pi_b)| + |S|$, the condition $\Delta > R$ forces $|\log(\pi_a/\pi_b)| > R - |S| \geq 0$: the prior log-odds must carry the entire excess on their own. So $\Delta > R$ *can* fire on a pinned pair, and when it does the certificate has been bought with prior mass rather than with evidence — a MAP that Corollary 7 has already said carries no discriminating support.

We state this because it is the honest limit of the result. The MAP condition is informative when the gate withholds little — which is when the phenomenon this paper is about is absent.

8 Exact divergences, and bounds

Proposition 9. $\mathrm{KL}(q\|p) = \log \mathbb{E}_q[w] - \mathbb{E}_q[\log w]$ *and* $\mathrm{KL}(p\|q) = \mathbb{E}_p[\log w] - \log \mathbb{E}_q[w]$.

Proof. $\sum_h q_h \log(q_h/p_h) = \sum_h q_h \log(\mathbb{E}_q[w]/w_h)$ by (3), and symmetrically for the second. □

The first is exactly the Jensen gap of $\log w$ under q. It is non-negative, and zero if and only if w is q-almost-surely constant — which is Proposition 6. So:

the gating distorts the posterior precisely to the extent that the validity factors vary across hypotheses, and by exactly the Jensen gap.

Proposition 10. *For every pair* a, b, $\left|\log(p_a/p_b) - \log(q_a/q_b)\right| \leq R$, *and* $\mathrm{TV}(p, q) \leq (w_{\max} - w_{\min})/(4\,\mathbb{E}_q[w])$.

Proof. The first is immediate from (4). For the second, $\mathrm{TV}(p, q) = \frac{1}{2}\sum_h q_h\,|w_h/\mathbb{E}_q[w] - 1| = \mathbb{E}_q|w - \mathbb{E}_q[w]|/(2\mathbb{E}_q[w])$, and the mean absolute deviation of a variable supported on $[w_{\min}, w_{\max}]$ is at most $(w_{\max} - w_{\min})/2$. □

The first bound is attained; the audit's worst slack over 20,000 instances is $+1.78 \times 10^{-15}$, equality to floating-point precision.

The second is attained too, and we can exhibit the witness. Take $q = (\frac{1}{2}, \frac{1}{2})$ and a two-point $w = (a, b)$. Then $\mathbb{E}_q[w] = (a + b)/2$, $p = (a, b)/(a + b)$, and

$$\mathrm{TV}(p, q) = \frac{|a - b|}{2(a + b)} = \frac{w_{\max} - w_{\min}}{4\,\mathbb{E}_q[w]}, \tag{11}$$

an identity rather than an inequality, on both ordering branches. So the bound **cannot be improved as a function of** $(w_{\max}, w_{\min}, \mathbb{E}_q[w])$. An earlier draft declined to claim this; the construction is elementary and we should have looked for it.

We still claim no more than that. Attainment on this witness says the constant 4 and the functional form are best possible *given those three arguments*; it says nothing about bounds using more information about w, and it is not a claim of priority — bounded-likelihood-ratio divergence bounds are an established subject [6].

9 Verification

Every proposition above is checked two ways, and the artifact is committed. Symbolically with a computer-algebra system at $n = 4$ (`simplify(lhs−rhs) = 0`, exact), and numerically over 20,000 random instances. Worst absolute deviations:

Result	Symbolic ($n = 4$)	Numeric worst deviation
Prop. 1 reweighting	exact	4.44×10^{-16}
Prop. 2 log-ratio	exact	1.78×10^{-15}
Prop. 3 pinned to prior	exact	8.88×10^{-16}
Prop. 4 gated score is a reweighting	—	5.55×10^{-16}
Prop. 4 shift = withheld contribution	—	3.55×10^{-15}
Cor. 5 pinned when all distinguishing gated	—	1.78×10^{-15}
Prop. 6 constant w	exact	—
Prop. 9 KL($q\|p$)	exact	1.33×10^{-15}
Prop. 9 KL($p\|q$)	exact	1.53×10^{-15}
Prop. 10 R bound	—	0 violations, slack $+1.78 \times 10^{-15}$
Prop. 10 TV bound	—	0 violations, slack -8.38×10^{-5}
Prop. 8 MAP condition	—	held in 2,721/2,721
Rem. 8 pinned pair never certifiable	—	0 violations, slack $+8.88 \times 10^{-16}$

Table 1: Verification. The symbolic pass is at $n = 4$; the hand proofs are for general n and the machine check is not.

10 An implemented instance

The condition in (9) is not hypothetical. A deployed cross-sensor diagnostic layer for small uncrewed aircraft maintains a naive-Bayes diagnosis over eight mutually exclusive causes, with a per-feature gate. For the pair (`nominal`, `gps_fault`) the symmetric difference is a single feature: the GNSS internal-physics consistency check. Its gate closes when the receiver reports no valid fix. So a fault that removes the fix removes the only term that could name it, and (9) is empty for the duration.

To test that reading without circularity, a checker was written that evaluates (9) directly from the committed signature table and a per-tick record of which features were active. It imports none of the production analysis — a test runs it in a fresh interpreter and asserts that no module of the diagnostic layer, and no part of its support analysis, appears in `sys.modules` afterwards. Its input carries no leader, no posterior and no support label. Against the production computation of the same quantity, over nine scenarios, 1500 ticks each and all 28 unordered hypothesis pairs:

What the agreement does and does not establish. It rules out that the collapse is an artefact of the production analysis pipeline: two implementations sharing no code reach the same verdict every time. It does **not** validate the signature table. Both evaluate LOW(a)△LOW(b) from the same committed configuration, so a table that misdescribes what the features measure would be misdescribed identically 378,000 times. Whether LOW(h) is faithful to the sensors is a separate question that no quantity of agreement can answer, and the structural argument above is downstream of it.

Tick-by-pair comparisons	378,000
Agreements	378,000
Disagreements	0
Cases in the one known disagreement class	0

Table 2: Independent checker against production. The two definitions differ in exactly one degenerate case, when a feature's consistency is exactly 0.5 and a structurally distinguishing term contributes zero numerically. That class was predicted before the comparison, counted separately, and did not occur.

What it found:

- (`nominal`, `gps_fault`) is the *only* pair among the 28 that ever collapses.
- It collapses for 76 ticks at the start of *every* scenario, including fault-free flight, because the gate is deliberately closed during a smoother warm-up. A structural property, not a fault.
- Under injected fix loss it collapses for a further 200 consecutive ticks — exactly the injected window — during which Corollary 5 says the weight ratio equals the prior ratio. (Corollary 5 rather than Proposition 3: the layer's gate is a skip, so it withholds a term that contributes differently to the two hypotheses, and the pinning arises from the exact cancellation in (8) rather than from $L_a = L_b$.)

A negative control on a current stack. The same predicate was run over 15,595 ticks from 20 trials on a current-generation autopilot in software-in-the-loop, in a campaign whose validity criterion was that the fix never dropped. Under our account the predicate must then report the pair distinguishable at every tick, and a single collapse would falsify it. It reports 0 collapses in 15,595 ticks. This is a control, not a replication of the fix-loss behaviour, and we do not present it as one: no fix-loss arm exists on that platform in our data.

The implemented consequence. The layer now reports a support state separately from the normalized hypothesis weight it emits. When $D(a, b) = \emptyset$ against a required competitor, the tick is emitted as *support lost for required information* rather than as eligibility — **at any weight, because no weight threshold enters the rule at all**. That is deliberate: conditioning identifiability on the weight would make it a function of the very score whose support is in question. The rule is likewise independent of freshness, so removing the freshness rule would not rescue it.

The absence of a threshold is visible in the data rather than only in the code. Across the ticks where the condition fires, the leader's weight ranges from 0.9691 to 0.9998; a rule gated at, say, 0.999 would have passed most of them.

11 What is new, stated plainly

Nothing in Sections 3–8 is new mathematics, and we do not want a reader to have to work that out. Table 3 is the audit.

What we claim is: the exact runtime-diagnostic formulation; the connection between hypothesis-dependent validity and posterior distortion stated as (4); the structural-support reading and its *static checkability* from a signature table; and the implemented counterexample of Section 10.

We claim no new information-theoretic result, no new Bayesian identity, and no sharper bound than what is already available.

Result	Status	Prior art
Prop. 1, 2	KNOWN	self-normalised importance weighting
Prop. 3, Cor. 7	KNOWN as algebra	—
Prop. 4, Cor. 5	KNOWN	reparameterisation of a log-linear score; one line of algebra
Prop. 6	KNOWN as algebra	the zero-loss corner of Prop. 1. It is *not* the coarsening-at-random condition [4]; the two are compared in Remark 3
Prop. 8	KNOWN	argmax stability under bounded perturbation
Prop. 9, 10	KNOWN	Jensen; bounded-support mean absolute deviation

Table 3: Novelty audit. Every proposition is labelled KNOWN.

12 Related work

The mechanism by which a fault removes its own observability is studied in discrete-event systems under state-dependent observation maps [3] and, for observation loss imposed from outside, under intermittent and permanent loss [1, 2] and general sensor failure [8]. We assert no priority over the general mechanism. Our setting differs in being a continuous-signal Bayesian layer in which applicability is hypothesis-relative and decided by the diagnostic system itself, so the condition is a property of its configuration rather than of an automaton.

The ignorability question is the coarsening-at-random condition [4], and the same question for a mechanism that depends on the latent state — ours does — is treated in latent-state missingness models, where the mechanism is a stochastic submodel estimated from data [7]. The difference we can state is that ours is deterministic and readable from the diagnostic system's own configuration, so it is decided rather than estimated; we assert no more than that, and in particular no claim that an estimated mechanism could not represent it. Which faults a given instrumentation can separate is the subject of structural isolability and sensor placement [5]; that literature answers the question our Section 6 raises but does not answer, namely what to instrument.

13 Limitations

1. **The baseline q is counterfactual and, in the measured case, not evaluable** (Remark 1). Propositions 9 and 10 therefore quantify a distance to a reference point rather than to anything the system computes. The pinning result does not depend on this and is stated in terms of p and the prior alone.

2. **Proposition 8 is vacuous in the regime we measure** (Remark 8): a pinned pair with equal prior mass satisfies $\Delta \leq R$ always, so the sufficient condition cannot fire. We have no MAP-stability result for the case the paper is about.

3. The symbolic verification is at $n = 4$. The hand proofs are general; the machine check is not. Propositions 4 and 8 and Remark 8 are checked numerically only.

4. The implemented instance is $n = 1$: one deployed layer, whose skip-style gate is a natural implementation choice but whose prevalence we do not establish. A structural-analysis FDI design would mark the fault set unisolable instead, and would not exhibit this.

5. The quantities the layer reports are model states, not calibrated probabilities, and we do not calibrate them. The results are invariant to the layer's tempering constant, which scales $\log q$ for every hypothesis alike.

6. No claim of operational harm is made, and none is supported by our data.

7. **Every bibliographic entry in this manuscript is unverified.** The authoring environment could not reach a publisher record. This is a submission blocker and is reported as one.

A separate body of work, not in this paper

This project also holds results on *anytime-valid* evidence — e-value calibration, blockwise e-processes and mission-level error control — with their own register at `docs/math/PROOF_STATUS.md`. That register contains six refutations, several of them counterexamples to constructions this project itself proposed, and four open items. It is a different body of work from the one above and is deliberately **not** part of this manuscript. We name it here so that a reader who encounters the register does not mistake its open items for gaps in the results proved above.

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